

SIDDHANT MIDHA

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EDUCATION

Princeton Quantum Initiative, Princeton University

Princeton, USA

Ph.D. in Quantum Science

2024-2029 (*expected*)

- Advisors: Prof. Dmitry Abanin, Prof. Sarang Gopalakrishnan
- GPA: 4.0/4.0
- Princeton First Year Fellowship in Natural Science & Engineering

2024-25

IIT Bombay

Mumbai, India

Bachelors in Electrical Engineering

2020 - 2024

- Minor in Physics
- Advisors: Prof. Sai V, Prof. Bhaskaran M, Prof. Kasturi S
- GPA: 9.77/10
- *Awards*

- Graduated with Honors, top 3% of class 2024
- Undergraduate Research Distinction 2024
- Dhananjay Fellowship 2023
- Chanakya Quantum Fellowship 2023
- Institute Academic Prize 2023
- Undergraduate Research Award 2023

RESEARCH INTERESTS

quantum error correction, quantum simulation, quantum metrology, tensor networks

PUBLICATIONS

1. SM, D. Abanin “Local autonomous inference machines for quantum LDPC codes”
[[arXiv:2609.38139](#)]
2. A. Basu, P. Kothari, SM “Sharp bounds on the ground state energy of the SYK model”
[[arXiv:2607.27185](#)]
3. SM, Y. Zhang, D. Malz, D. Abanin, S. Gopalakrishnan “Algorithmic locality via provable convergence in quantum tensor networks ” [[arXiv:2604.21919](#)]
4. SM^{*}, G. Sommers^{*}, J. Tindall, D. Abanin “Belief Propagation & Tensor Network Expansions for Many-Body Quantum Systems: Rigorous Results and Fundamental Limits”
[[arXiv:2604.03228](#)]
5. SM^{*}, Y. Zhang^{*} “Beyond Belief Propagation: Cluster-Corrected Tensor Network Contraction with Exponential Convergence”
[[arXiv:2510.02290](#), [PRX Quantum 7, 033010](#), [TQC 2026](#)]
6. SM, S. Gopalakrishnan “Metrology of open quantum systems from emitted radiation”
[[arXiv:2504.13815](#), [PRA Letter 112, L060601](#)]
7. K. Agaram, SM, V. Garg “Preparing stabilizer states via path-aware reinforcement learning” [[Neurips Machine Learning for Physical Sciences](#), [APS Global Summit 2025](#)]
8. A. Arora^{*}, SM^{*}, A. Zyuzin, P. Hakonen, B. Muralidharan, “Steady-state dynamics and correlations in quantum-dot Cooper pair splitters”
[[arXiv:2406.06053](#), [npj Quantum Information 11, 40 \(2025\)](#), [APS March Meeting 2024](#)]
9. SM, K. Jana, B. Muralidharan, “Are Symmetry Protected Topological Phases Immune to Dephasing? — A topological electronics perspective” [[arXiv:2305.11149](#), [J. Phys. D: Appl. Phys. 57 135308](#)]

10. SM, M. Parashar, A. Bhatla, D. Broadway, J.P. Tetienne, K. Saha,
“Optimized Current Density Reconstruction from Widefield Quantum Diamond Magnetic Field Maps” [[arXiv:2402.17781](#), [Phys. Rev. Applied 22, 014015](#)]
11. SM, R. Singh, K. Gharavi, J. Baugh, B. Muralidharan,
“On the microscopies of proximity effects in one-dimensional superconducting hybrid systems” [[arXiv:2411.12733](#)]

*denotes equal contribution

AWARDS

- **DAAD-WISE Fellowship**, German Academic Exchange Service 2023
- **MITACS Globalink Fellowship**, Canada (unavailed) 2023
- **Aalto Science Fellowship**, Aalto University 2022
- **All India Rank 150**, the JEE Advance exam (top 0.09%) 2020
- **KVPY Award**, Govt. of India (unavailed) 2020

TALKS

1. Beyond Belief Propagation: Cluster-Corrected Tensor Network Contraction
 - (a) Google Quantum AI Tensor Network Meeting (online) ♦ 2025
 - (b) Thompson group meeting, Princeton, USA ♦ 2025
 - (c) International Conference on Theory of Quantum Computation, Communication and Cryptography (TQC) 2026
2. Taming the many-body Hilbert Space: Tensor Networks and Belief Propagation
 - (a) Physics Department Seminar, IIT Bombay, India ♦ 2026
 - (b) Algorithms and Complexity Seminar, Princeton, USA ♦ 2026
 - (c) ITensor Seminar, Flatiron Institute, NYC, USA ♦ 2026
 - (d) Max-Planck Quantum Theory Group Seminar ♦(online) 2026
3. Algorithmic locality in quantum tensor networks
 - (a) Interplay between neural and tensor networks, Grenoble, France 2026
4. Local autonomous inference machines for quantum LDPC codes
 - (a) Princeton Quantum Initiative Group Meeting, Princeton, NJ 2026
5. Quantum metrology of continuously monitored open systems
 - (a) Blackboard seminar, UC San Diego, USA 2025
 - (b) Quantum Engineering Seminar, IIT Bombay, India ♦ 2026
 - (c) Relativistic Quantum Information Online Circuit 2026
6. Solid-state entanglement generation with Cooper-pair splitters
 - (a) APS march meeting, Minneapolis, MN, USA 2024

♦ denotes invited talk

EXPERIENCE

- Undergraduate researcher**
TU Munich & MPQ, Germany. 2023.05 - 2023.07
- Undergraduate researcher**
Aalto University, Finland. 2022.05 - 2022.07
- Research internship (virtual)**
QIntern. 2022.05 - 2022.07

MEDIA

[Quantum Zeitgeist](#) “Cluster-corrected Tensor Network Contraction Achieves Exponential Convergence with Rigorous Error Bounds”

TEACHING

1. Advanced Quantum Mechanics, Princeton 2026
2. Information theory and applications, Princeton 2025

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| 3. Quantum information and computing, IIT Bombay | 2024 |
| 4. Calculus I, IIT Bombay | 2022 |
| 5. Complex Analysis, IIT Bombay | 2022 |
| 6. Differential equations, IIT Bombay | 2021 |
| 7. Linear algebra, IIT Bombay | 2021 |
| 8. Calculus II, IIT Bombay | 2021, 2022 |

SKILLS

Programming: Python, Julia

Libraries: QuTiP, Qiskit, ITensor, Stim, PyTorch

Analysis: Mathematical proofs, optimization, information theory, statistical & data analysis

MENTORSHIP

Mentored Krishna Agaram, junior undergraduate at IIT Bombay

Institute Student Mentor: to twelve freshmen at IIT Bombay

Department Academic Mentor: to eight sophomores at IIT Bombay

Quantum outreach

1. Mentored 12+ junior undergraduates in quantum computing research at IIT Bombay
[[Learning \(with\) quantum computers](#), [Quantum machine learning](#)]
2. Princeton Undergrads in Quantum Talk [A conversation on quantum information](#)

SERVICE

Journal referee Physical Review Letters, Physical Review A

Conference referee for NeurIPS machine learning for physical sciences 2024, 2025